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Full Course Assessment

Review Learning Objectives

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Assignment details

Due
Dec 27, 11:59 PM IST

Attempts
Unlimited

Submitted
Dec 11, 4:43 AM IST

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1. Which of the following statements correctly describe the concept of superposition in qubits? **1 / 1 point**

☐ Superposition is a fundamental property of classical bits

☑ Superposition is represented using Dirac notation

Correct
Correct! Dirac notation is used to represent the quantum states in superposition.

☐ Superposition prevents qubits from interacting with each other

☑ A qubit can exist in multi[illegible]ntil measured

Correct
Correct! Superposition allows a qubit to be in a combination of states simultaneously.

☑ Superposition enables a qubit to perform parallel processing tasks

Correct
Correct! Superposition allows quantum computers to perform many calculations at once, leading to parallel processing capabilities.

2. Which of the following methods are used for generating and retaining qubits in quantum computers? **1 / 1 point**

☑ Quantum dots

Correct
Correct! Quantum dots are another method for creating qubits and are used in various quantum computing implementations.

☑ Ion traps

Correct
Correct! Ion traps are a proven method for generating and retaining qubits with high coherence times.

☑ Superconducting qubits

Correct
Correct! Superconducting qubits are widely used due to their coherence times and scalability.

☐ Phase-change memory

☐ Classical transistors

3. Calculate the dot product of the following two matrices and identify its significance in quantum computing: **1 / 1 point**

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$$

.

● Dot product is 14; it represents the inner product in the quantum Hilbert space.

○ Dot product is 14; it represents the probability amplitude of quantum states.

○ Dot product is 10; it represents the probability amplitude of quantum states.

○ Dot product is 10; it represents the inner product in the quantum Hilbert space.

Correct
Correct! The dot product is 14, and it represents the inner product in the quantum Hilbert space.

4. What is the primary function of the Pauli X gate in a quantum circuit? **1 / 1 point**

○ To apply a phase shift to a qubit

○ To entangle two qubits

○ To measure the state of a qubit in the computational basis

● To flip the state of a qubit from $|0\rangle$ to $|1\rangle$ or vice versa

Correct
Correct! The Pauli X gate is also known as the Pauli-X gate and it flips the state of a qubit.

5. What effect does applying the Pauli Y gate have on a qubit's state vector? **1 / 1 point**

○ It flips the qubit state from $|0\rangle$ to $|1\rangle$ or from $|1\rangle$ to $|0\rangle$.

○ It entangles the qubit with another qubit in the circuit.

○ It measures the state of the qubit.

● It performs a rotation of the state vector around the Y axis by 90 degrees.

Correct
Correct! The Pauli Y gate rotates the state vector around the Y axis by 90 degrees, which is essential for certain quantum operations.

6. Which of the following statements accurately describe the polygate in comparison to X and Y gates? **0.75 / 1 point**

☐ Polygate measures qubits in the Z basis.

☐ Polygate can be represented as a combination of X and Y gates.

☑ Polygate can perform rotations around an arbitrary axis.

Correct
Correct! A polygate can perform rotations around any axis, unlike X and Y gates.

☐ Polygate is used to entangle qubits.

You didn't select all the correct answers

7. Which of the following steps are involved in the process of creating the X gate using Hadamard (H) and C gates? **1 / 1 point**

☑ Apply a Hadamard gate to the qubit.

Correct
Correct! The Hadamard gate is indeed applied to the qubit as part of the process of creating the X gate.

☐ Apply a phase flip gate.

☑ Apply a C gate to the qubit.

Correct
Correct! The C gate is also applied to the qubit in the process of creating the X gate.

☐ Measure the qubit in the computational basis.

☑ Apply another Hadamard gate after the C gate.

Correct
Correct! After applying the C gate, another Hadamard gate is applied to complete the process of creating the X gate.

☐ Reset the qubit to the ground state.

8. Select all statements that accurately describe the R Phi gate and RC gate in quantum computing. **1 / 1 point**

☐ The RC gate is mainly used for entangling qubits.

☐ Both R Phi and RC gates are used to reset qubit states.

☑ The R Phi gate allows for arbitrary rotations around the z-axis.

Correct
Correct! The R Phi gate is used for arbitrary rotations around the z-axis.

☑ The RC gate is a combination of rotation and control operations.

Correct
Correct! The RC gate combines rotation with control operations.

☐ The R Phi gate can only perform rotations around the x-axis.

9. Which of the following statements correctly describes the properties of the T gate in relation to the R Phi gate? **1 / 1 point**

○ The T gate is a special case of the Hadamard gate where

$$\phi = \frac{\pi}{4}$$

.

○ The T gate is a specific case of the R Phi gate with

$$\phi = \frac{\pi}{2}$$

and applies a Hadamard transformation.

● The T gate is a specific case of the R Phi gate with

$$\phi = \frac{\pi}{4}$$

and applies a phase shift.

○ The T gate is unrelated to the R Phi gate and serves a different purpose in quantum circuits.

Correct
Great job! The T gate is indeed a specific case of the R Phi gate with a phase shift of

$$\frac{\pi}{4}$$

. This property is essential for various quantum algorithms.

10. Select all correct descriptions of the U1, U2, and I gates. **0.8 / 1 point**

☐ The U2 gate is a rotation around the Y axis followed by a phase shift around the Z axis, with no rotation around the X axis.

☐ The U1 gate is the same as the U3 gate with $\theta = \pi/2$.

☑ The U1 gate applies a phase rotation to a single qubit.

Correct
Correct! The U1 gate applies a phase shift.

☑ The I gate is the identity gate, which leaves the qubit state unchanged.

Correct
Correct! The I gate does not alter the qubit state.

☐ The U2 gate is equivalent to applying two consecutive U1 gates.

You didn't select all the correct answers

11. Which of the following steps are required to create a sample quantum circuit with multiple qubits in Qiskit? Select all that apply. **0.8 / 1 point**

☑ Apply quantum gates to the qubits

Correct
Good job! Applying quantum gates is a crucial part of building quantum circuits.

☐ Initialize all qubits to the state

$$|1$$

\|

☑ Import the necessary Qiskit libraries

Correct
Correct! Importing Qiskit libraries is essential for building quantum circuits.

☐ Measure the quantum state of each qubit

☐ Set up a classical control system

You didn't select all the correct answers

12. Which of the following steps are necessary to implement a quantum circuit in Qiskit and interpret the resulting output? **0.8 / 1 point**

☑ Initialize the quantum circuit with the desired number of qubits and classical bits

Correct
Correct! Initializing the quantum circuit is the first step in creating a quantum circuit in Qiskit.

☐ Measure the qubits and store the results in the classical bits

☐ Initialize all qubits to the

$$|1\rangle$$

state before applying gates

☑ Apply quantum gates to the qubits to manipulate their states

Correct
Correct! Applying quantum gates is essential for manipulating the state of the qubits in the circuit.

☐ Run the circuit on a classical computer to get the final output

You didn't select all the correct answers

13. Which of the following statements about the truth table of the CNOT gate are correct? Select all that apply. **1 / 1 point**

☑ If the control qubit is

$$|0\rangle$$

and the target qubit is

$$|0\rangle$$

, the output is

$$|00\rangle$$

.

Correct
Correct! When the control qubit is

$$|0\rangle$$

and the target qubit is

$$|0\rangle$$

, the output remains

$$|00\rangle$$

. This follows the CNOT gate's truth table.

☐ If the control qubit is

$$|1\rangle$$

and the target qubit is

$$|1\rangle$$

, the output is

$$|11\rangle$$

.

☑ If the control qubit is

$$|1\rangle$$

and the target qubit is

$$|1\rangle$$

, the output is

$$|10\rangle$$

.

Correct
Correct! When the control qubit is

$$|1\rangle$$

and the target qubit is

$$|1\rangle$$

, the output is

$$|10\rangle$$

. This follows the CNOT gate's truth table.

☑ If the control qubit is

$$|1\rangle$$

and the target qubit is

$$|0\rangle$$

, the output is

$$|11\rangle$$

.

Correct
Correct! When the control qubit is

$$|1\rangle$$

and the target qubit is

$$|0\rangle$$

, the output is

$$|11\rangle$$

. This is consistent with the CNOT gate's truth table.

☐ If the control qubit is

$$|1\rangle$$

and the target qubit is

$$|0\rangle$$

, the output is

$$|10\rangle$$

.

☑ If the control qubit is

$$|0\rangle$$

and the target qubit is

$$|1\rangle$$

, the output is

$$|01\rangle$$

.

Correct
Correct! When the control qubit is

$$|0\rangle$$

and the target qubit is

$$|1\rangle$$

, the output remains

$$|01\rangle$$

. This is directly deduced from the CNOT gate's truth table.

14. What is the effect of applying the X gate to a qubit initially in state **1 / 1 point**

$$|0\rangle$$

?

○ The qubit's state becomes undefined

○ The qubit remains in state

$$|0\rangle$$

● The qubit transitions to state

$$|1\rangle$$

○ The qubit enters into a superposition state

Correct
Correct! The X gate flips the qubit from state

$$|0\rangle$$

to state

$$|1\rangle$$

.

15. Which of the following statements accurately describe the controlled insert gate and its properties? **0.8 / 1 point**

☐ The controlled insert gate is a single-qubit gate.

☑ The controlled insert gate can be represented using a matrix.

Correct
Correct! The controlled insert gate has a corresponding matrix representation.

☐ The controlled insert gate is used to create entanglement between qubits.

☐ The state vector of the system remains unchanged after applying the controlled insert gate.

☑ The controlled insert gate alters the state of the target qubit based on the state of the control qubit.

Correct
Correct! The controlled insert gate changes the state of the target qubit depending on the control qubit's state.

You didn't select all the correct answers

16. Which of the following statements are true about the unitary matrix representation of the controlled-Y (CY) gate? **0.75 / 1 point**

☐ The matrix swaps the qubit states conditionally based on the control qubit.

☐ The matrix includes a block that resembles the identity matrix for certain qubit states.

☑ The matrix has dimensions of 4x4.

Correct
Correct! The unitary matrix of a two-qubit gate like the CY gate has dimensions of 4x4.

☐ The matrix elements are all real numbers.

You didn't select all the correct answers

17. What is the primary role of a swap gate in a quantum circuit? **1 / 1 point**

● To exchange the states of two qubits

○ To measure the state of a qubit

○ To entangle two qubits

○ To create a superposition state

Correct
Correct! A swap gate is used to exchange the states between two qubits.

18. What is the primary function of the Toffoli gate in quantum computing? **1 / 1 point**

○ To measure the state of a qubit.

○ To create a superposition of states.

● To perform a controlled-controlled-not operation.

○ To entangle two qubits.

Correct
Correct! The Toffoli gate is also known as the controlled-controlled-not gate.

19. Which gate would you use in Qiskit to create superposition in a qubit? **1 / 1 point**

○ Z gate

○ X gate

● H gate

○ CNOT gate

Correct
Correct! The H gate, also known as the Hadamard gate, is used to create superposition in a qubit.

20. What is the primary purpose of quantum key distribution (QKD)? **1 / 1 point**

○ To secure communications by entangling the communicating parties.

○ To replace classical encryption methods entirely with quantum algorithms.

● To securely transfer encryption keys using the principles of quantum mechanics.

○ To increase the speed of classical encryption algorithms.

Correct
Correct! The primary purpose of QKD is to transfer encryption keys securely using quantum mechanics, ensuring that any eavesdropping attempts are detectable.